

Cirrhosis Patient

SURVIVAL PREDICTION

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Introduction

Cirrhosis is a serious liver disease that progresses through several stages, and identifying a patient's current stage can be challenging because symptoms and lab results often vary widely. This study aims to build a model that helps classify the stage of cirrhosis using key clinical signs and laboratory measurements, while also highlighting the factors that most strongly influence disease progression. The Cirrhosis Patient Survival Prediction dataset includes patient demographics, clinical features such as ascites, hepatomegaly, spider angiomas, and edema, as well as lab values like bilirubin, albumin, SGOT, and prothrombin time. By analyzing these variables, the study supports SDG 3 (Good Health and Well-Being) through earlier detection of disease severity and improved decision-making in patient care.

Data Understanding

	Age	Sex	Ascites	Hepatomegaly	Spiders	Edema	Bilirubin	Albumin	Copper	Alk_Phos	SGOT	Platelets	Prothrombin	Stage
0	21464.0	F	Y	Y	Y	Y	14.5	2.60	156.0	1718.0	137.95	190.0	12.2	4.0

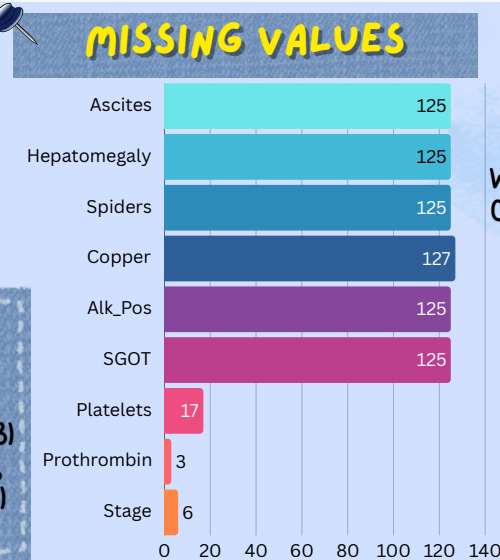
ROW : 620
COLUMN : 20

DUPLICATES : 2

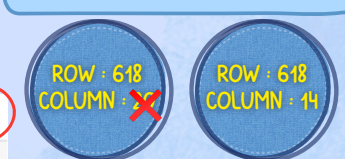
THIS DATASET CONTAINS REPEATED RECORDS THAT SHOULD BE REMOVED TO ENSURE RELIABLE ANALYSIS.

INCONSISTENT & OUTLIERS

SEX LABELS DETECTED: 'F', 'M', 'FEMALE', 'MALE', 'MALE'
ASCITES LABELS DETECTED: 'YES', 'NO', 'Y', 'N'.
BILIRUBIN (82), ALK_PHOS (48), PLATELETS (45), AND SGOT (33) HAVE THE MOST OUTLIERS. COPPER (16), PROTHROMBIN (24), AND ALBUMIN (12) SHOW MODERATE COUNTS, WHILE AGE (6) HAS FEW AND STAGE (0) HAS NONE.



DROP IRRELEVANT COLUMNS



Data Preparation

Stage 1	21	Stage 1	21
Stage 2	97	Stage 2	97
Stage 3	311	Stage 3	311
Stage 4	183	Stage 4	183
Nan	6		

TRAIN

70%

30%

DUPLICATE ROWS
WERE REMOVED TO ENSURE CLEAN AND UNBIASED DATA

OUTLIERS
APPLIED LOG1P TRANSFORMATION TO SGOT, BILIRUBIN, AND PLATELETS TO REDUCE THE EFFECT OF EXTREME VALUES

NORMALIZATION

USED ROBUST SCALER TO SCALE SGOT, COPPER, BILIRUBIN, AND PLATELETS BECAUSE THESE FEATURES CONTAIN MANY OUTLIERS AND THIS METHOD IS LESS SENSITIVE TO EXTREME VALUES.

MISSING VALUES
NUMERIC FEATURES: FILLED USING MEDIAN

MISSING VALUES
CATEGORICAL FEATURES: FILLED USING MODE

INCONSISTENT SEX LABELS WERE UNIFIED BY CONVERTING 'MALE' → 'M', 'FEMALE' → 'F', 'MALE' → 'M'

INCONSISTENT ASCITES LABELS WERE HARMONIZED BY CONVERTING 'YES' → 'Y', 'NO' → 'N'

BINARY ENCODE

Sex			
M	1	F	0
Ascites			
Y	1	N	0
Hepatomegaly			
Y	1	N	0
Spiders			
Y	1	N	0

LABEL ENCODE

Edema	
N	0
S	1
Y	2

Machine Learning Modelling



In this project, we applied five models: Logistic Regression, Random Forest, XGBoost, CatBoost, and Light GBM. Based on the evaluation metrics obtained, we focused on **recall** as the key indicator for determining the best-performing model. Among the five models, **Light GBM** achieved the highest scores, followed by **XG Boosting**. Therefore, these top two models will be further tuned to evaluate whether their performance can be improved even more.

Light GBM

Before Tuning

Stage	Recall
1	0.25
2	0.26
3	0.75
4	0.74

Hyperparameter Tuning

Stage	Recall
1	0.25
2	0.19
3	0.74
4	0.72

Threshold Tuning

Stage	Recall
1	0.25
2	0.23
3	0.73
4	0.81

XG Boosting

Before Tuning

Stage	Recall
1	0.25
2	0.19
3	0.77
4	0.70

Hyperparameter Tuning

Stage	Recall
1	0.25
2	0.23
3	0.73
4	0.68

Threshold Tuning

Stage	Recall
1	0.25
2	0.26
3	0.72
4	0.70

- Stage 1 (Early Stage): LightGBM achieves higher recall after threshold tuning, indicating better sensitivity in detecting early cases, while XGBoost shows relatively stable recall with limited improvement.
- Stage 2: Recall improves more clearly in LightGBM after threshold tuning, meaning intermediate cases are better captured, whereas XGBoost gains mainly from hyperparameter tuning.
- Stage 3: Both models reach their highest recall during hyperparameter tuning, suggesting that model structure plays a more important role than threshold adjustment at this stage.
- Stage 4 (Late Stage): LightGBM shows the strongest recall after threshold tuning, indicating improved detection of severe cases, while XGBoost exhibits smaller recall gains.

Based on recall evaluation, LightGBM performs best after threshold tuning, particularly in improving detection of higher stages. In contrast, XGBoost achieves its optimal performance through hyperparameter tuning, showing more stable recall without relying heavily on threshold adjustment. This indicates that LightGBM is more sensitive to threshold changes, while XGBoost benefits more from model structure optimization.

Conclusion

This study demonstrates that machine learning models can effectively classify cirrhosis stages using a combination of clinical symptoms and laboratory indicators. Among the evaluated models, LightGBM showed the most notable improvement, especially after threshold tuning, particularly by achieving higher recall in early and late stages, which is critical for timely intervention. These results support SDG 3 (Good Health and Well-Being) by enhancing disease detection and improving clinical decision-making, ultimately helping prevent disease progression and reduce severe health outcomes.

Access our code here



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